

JALAJ SANJAY MEHTA

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Materials Science & Engineering · Computer Science · Product & Commercialization

EDUCATION

University of Pennsylvania | M.S.E. Materials Science & Engineering | GPA: 3.85 2025–2026

Relevant Coursework:

Nanofabrication & Nanocharacterization - hands-on cleanroom fabrication (optical/e-beam lithography, PVD, etching, directed self-assembly, quantum dot characterization, graphene devices); Commercialization of Academic Science (Wharton/Mack Institute) - customer discovery, go-to-market strategy, and competitive analysis for early-stage Penn research; Product Management - product life-cycle strategy, design thinking, agile methods, and product-portfolio metrics

Columbia University | B.S. Materials Science & Engineering | Minor: Computer Science | GPA: 3.65 2021–2025
Dean's List (all eligible semesters) · Bonomi Scholar (Summer 2022) · Principled Action Leadership & Engagement Award

PROFESSIONAL EXPERIENCE

August | Product Developer – Legal AI Systems Dec 2025–Present

- Review user and AI assistant interaction data to surface bugs; translate findings into structured feedback for engineering team
- Advise on feature prioritization and programming direction, bridging user needs and technical implementation
- Train cross-functional teammates, improving product-development velocity and workflow consistency

Northrop Grumman Mission Systems | College Technical Intern – Transistor Process Control Jun–Aug 2024

- Built a Python application with a UX to ingest bulk transistor data across file types, deriving I–V curves, capacitance, conductance, and power parameters from raw measurements
- Integrated a virtual-source transistor model into Python, validated against measured data, and extended it with carrier freeze-out corrections for cryogenic operation
- Benchmarked transistors from multiple manufacturers using lab test equipment, delivering process-control insights to the team

Columbia University | Undergraduate Student Life – Community Events Coordinator, NSOP May–Sep 2023

- Designed and led the full orientation schedule for Columbia's incoming class; trained hundreds of Orientation Leaders and Crew Captains, and authored event minute-by-minutes
- Collaborated with senior administration to develop Columbia-specific programming; served as primary external contact for community-building events

RESEARCH EXPERIENCE

Columbia MSE Dept — Senior Design | Research Assistant Oct 2024–May 2025

- Used qNMR and DSC to construct phase diagrams of organic liquid electrolytes, building full electrochemical cells from scratch as part of next-generation lightweight, high-energy-density battery research
- Built a KNN Machine Learning Model to predict electrolyte phase diagrams based on collected data

Columbia ChemE / Brookhaven National Lab | Research Assistant & Visiting Researcher Feb 2022–May 2024

- Established a novel metallization procedure for DNA crystal lattices via iron-oxide infiltration; synthesized DNA crystals and characterized mechanical properties via nanoindentation
- Employed SEM, TEM, FIB, optical microscopy, XRD, and EDX for structural and compositional characterization; developed algorithms to extract mechanical properties from raw nanoindentation data

National Energy Technology Lab (DOE Mickey Leland Fellowship) | Undergraduate Fellow Jun–Aug 2023

- Authored a Python program for image processing of TEM micrographs, implementing 3D matrix mathematics to compute crystal grain misorientation angles and axes; packaged as a Tkinter executable

SUNY Stony Brook – Garcia Research Program | Student Researcher Jul 2019–Mar 2021

- Developed a biodegradable PVA-based flame-retardant hydrogel; modeled UiO-66 MOF carbon-capture properties via the Lattice Boltzmann method
- Published in ACS Biomacromolecules (2021); presented at the Fall 2019 MRS Conference, Boston

LEADERSHIP

Columbia University Senate & Engineering Student Council | SEAS Undergraduate Senator Oct 2023–May 2025

- Represented engineering undergraduates in Columbia's primary shared-governance body; served on Student Affairs, Rules of University Conduct, and Structures & Operations committees
- Collaborated with administration, faculty, and peer councils to advance student-facing proposals and host community programming

SKILLS & BACKGROUND

Programming & Tools: Python (advanced – data pipelines, image processing, GUI, scientific computing), Java, C, SQL, Arduino, LaTeX, ImageJ, Onshape, Ultimaker Cura, Microsoft Excel & PowerPoint

Characterization & Fabrication: SEM, TEM, FIB, XRD, EDX, optical microscopy, nanoindentation, qNMR, DSC, cleanroom nanofabrication, electrochemical cell assembly

Product & Business: Customer discovery, go-to-market strategy, product roadmapping, design thinking, agile workflows, user feedback analysis, competitive analysis, technology commercialization

Languages: English (native), Gujarati (native), Spanish (conversational)

PUBLICATIONS & PRESENTATIONS

Mehta et al., "Synthesis of an Effective Flame-Retardant Hydrogel for Skin Protection Using Xanthan Gum and RDP-Coated Starch," **ACS Biomacromolecules**, 2021

Presenter, Fall 2019 Materials Research Society Conference, Boston – Flame Retardancy of Biodegradable PVA Hydrogels

Presenter, U.S. Department of Energy Symposium, Washington, D.C., 2023 – TEM Image Processing for Crystal Grain Analysis